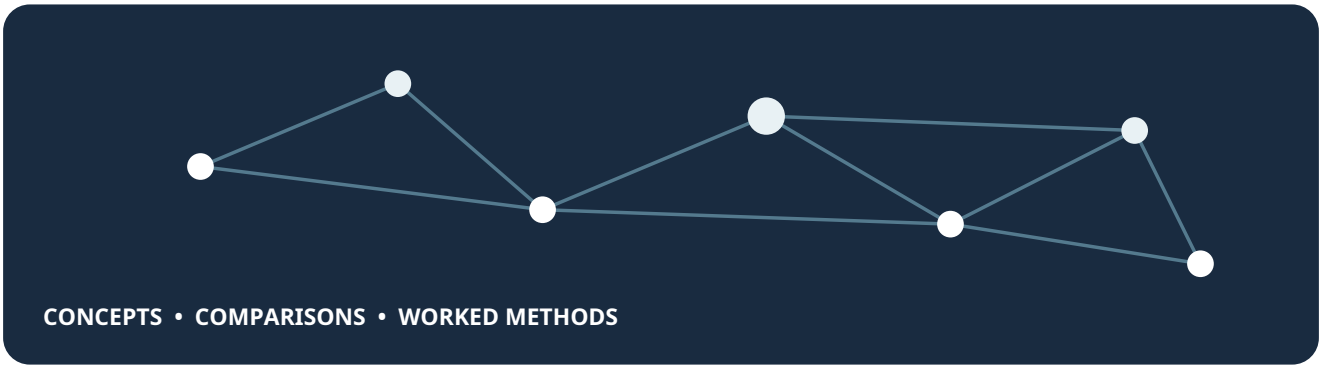
GTU / BE COMPUTER ENGINEERING / SEMESTER 5

COMPUTER NETWORKS

Viva Cheat Sheet



Built strictly around the five official syllabus chapters. Repeated written-exam topics receive extra emphasis; new-syllabus topics are kept concise but are not ignored. Historical frequency is a guide, not a guarantee for the viva.

Course code	University	Use
BE05000171	Gujarat Technological University	Fast pre-viva revision

A revision reference, not a replacement for your lecturer's instructions or the official syllabus.

START HERE

One subject. Five chapters. Clear priorities.

Use this page to decide what to revise first. Then jump to the matching chapter.

Whole-subject overview
Computer Networks explains how devices exchange data through layered protocols, addressing, routing, applications, and security.

The five-chapter map

CHAPTER	ONE-LINE OVERVIEW	REVISION FOCUS
1 Introduction	Network models, media, topologies, switching, Ethernet and IP versions.	OSI/TCP-IP; topologies; switching
2 Data Link + Network	Frames, error checks, MAC access, IP subnetting, switching and routing.	CRC; subnetting; DVR; Dijkstra; MAC
3 Transport + Application	TCP/UDP, flow and congestion control, DNS, web, email and sockets.	HTTP; DNS; GBN/SR; sockets; TCP/UDP
4 Wireless, IoT + Cloud	Wi-Fi, Bluetooth, 5G, MQTT/CoAP, cloud, VMs and containers.	Wi-Fi; 5G; MQTT vs CoAP; VM vs container
5 Security + Emerging	Firewalls, VPN, IDS/IPS, cryptography, TLS, SDN and zero trust.	SDN; firewall; VPN; crypto; IDS/IPS

High-priority revision map

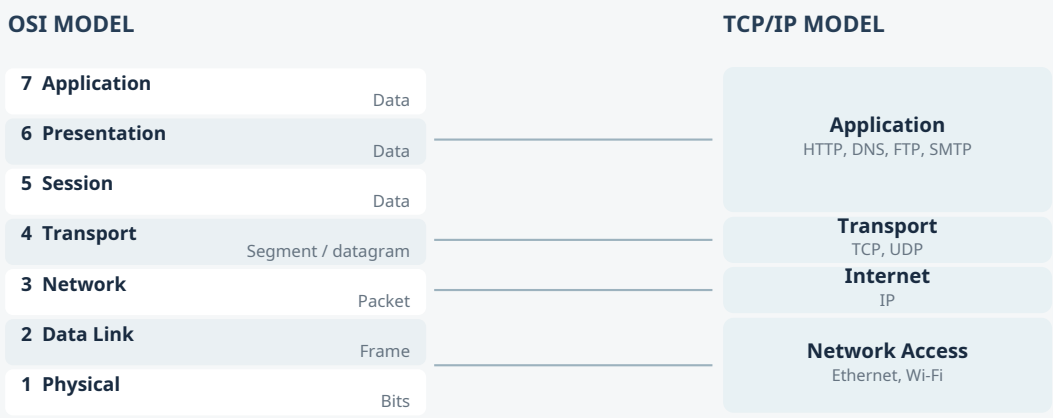
TOPIC	CH.	PYQ SIGNAL
CRC / codeword generation	2	M.IMP: appeared in 9 of 11 analyzed papers
Distance Vector Routing	2	M.IMP: appeared in 9 of 11
HTTP request/response + connection behaviour	3	M.IMP: appeared in 9 of 11
DNS resolution	3	M.IMP: appeared in 8 of 11
OSI/TCP-IP models	1	M.IMP: appeared in 8 of 11
Socket programming; GBN/SR; subnetting; Dijkstra	2-3	M.IMP: each appeared in 7-8 of 11
Wi-Fi, 5G, IoT/cloud, SDN/security	4-5	Current syllabus topics; no historical PYQ coverage in supplied analysis

PYQ counts come from the supplied analysis of 11 papers (Winter 2020 to Winter 2025). They refer to written exams and do not guarantee viva questions.

CHAPTER 1 / INTRODUCTION

OSI and TCP/IP models

Learn the layer order, one main job per layer, and how the two models map together.



OSI 7/6/5 -> Application; 4 -> Transport; 3 -> Internet; 2/1 -> Network Access.

Layer-by-layer recall

OSI LAYER	CORE JOB	EXAMPLE / PDU
Application	Network services used by applications.	HTTP, DNS; data
Presentation	Data format and representation.	Encoding / encryption; data
Session	Organises application dialog.	Session control; data
Transport	End-to-end process delivery; reliability when provided.	TCP / UDP; segment / datagram
Network	Logical addressing and routing between networks.	IP; packet
Data Link	Local frame delivery and MAC addressing.	Ethernet / Wi-Fi; frame
Physical	Transmits raw bits as signals.	Bits

Memory order: Please Do Not Throw Sausage Pizza Away = Physical, Data Link, Network, Transport, Session, Presentation, Application. Read bottom-up for sending data.

Viva answer: OSI is a seven-layer reference model. TCP/IP is the practical protocol architecture commonly shown with four layers: Network Access, Internet, Transport, and Application.

CHAPTER 1 / INTRODUCTION

Topologies, transmission and switching

Know the shape, the trade-off, and where each concept fits.



Concept sketches, not physical scale.

Topology trade-offs

TOPOLOGY	ADVANTAGE	LIMITATION
Star	Easy to add, isolate and troubleshoot a link.	Central switch/hub failure affects the LAN.
Bus	Simple, less cabling in a small setup.	Backbone fault can disrupt the segment.
Ring	Predictable path in a ring design.	A break can disrupt a basic single ring.
Mesh	Redundant paths; strong resilience.	More links, ports and cost.
Tree	Scalable hierarchical layout.	Upper-level failure can affect branches.

Two more high-value distinctions

CONCEPT	REMEMBER THIS
Transmission media	Guided: copper or fibre carries signals through a physical path. Unguided: radio, microwave or infrared travels through space.
Circuit vs packet switching	Circuit: reserves a path/resources for a session. Packet: divides data into packets that share links; packets may take different routes.
Ethernet and LAN	Ethernet is a common wired LAN technology (IEEE 802.3). Ethernet frames carry source and destination MAC addresses.
IPv4 vs IPv6	IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses. IPv6 greatly expands address space and uses hexadecimal colon notation.

Do not confuse a topology (how devices/links are arranged) with a protocol (the rules used to communicate).

CHAPTER 2 / DATA LINK + NETWORK

CRC, parity and framing

The CRC method is a top revision priority. Practise the order, not just the definition.

M.IMP | CRC / codeword generation appeared in 9 of the 11 papers in your supplied analysis.

CRC in six steps

- 1. Write the data bits and generator bit pattern.
- 2. If the generator has r + 1 bits, append r zeros to the data.
- 3. Divide using modulo-2 arithmetic. Use XOR instead of normal subtraction; do not carry or borrow.
- 4. The final r bits are the remainder.
- 5. Append that remainder to the original data to form the transmitted codeword.
- 6. The receiver divides the received codeword by the same generator. A zero remainder means no error was detected, not that every possible error is impossible.

Worked example

STEP	VALUE
Data	1101
Generator	1011 (4 bits, so append 3 zeros)
Padded data	1101000
Modulo-2 division remainder	001
Transmitted codeword	1101 + 001 = 1101001
Receiver check	1101001 divided by 1011 gives remainder 000

XOR division trace (working bits after each pass)

PASS	WORKING BITS
Start with padded data	1101000
XOR generator at first 1	0110000
Next XOR pass	0011100
Next XOR pass	0001010
Final XOR pass	0000001 -> remainder 001

Framing and error checks

METHOD	FAST RECALL
Parity	Adds a parity bit to make the count of 1s even or odd. Simple parity can detect many errors, but cannot correct them.
Bit stuffing	A sender inserts a 0 after a defined run of 1s so data is not mistaken for a frame delimiter. The receiver removes the stuffed bit.
Framing	Defines where one data-link frame starts and ends, so the receiver can separate consecutive frames.

CHAPTER 2 / DATA LINK + NETWORK

IPv4 subnetting and CIDR

Find the prefix, block size, network, broadcast and usable range in a fixed order.

M.IMP | Subnetting, subnet masks, classless addressing and VLSM appeared in 7 of 11 analyzed papers.

The method

- 1. Read the prefix, for example /26 means 26 network bits and 6 host bits.
- 2. Convert the prefix to a subnet mask. /26 = 255.255.255.192.
- 3. Block size in the changing octet = 256 - mask value. Here: 256 - 192 = 64.
- 4. Identify the block containing the IP address. That block start is the network address; the next block start minus 1 is the broadcast address.
- 5. For a conventional IPv4 subnet, usable hosts are between network and broadcast addresses.

Worked example: 192.168.1.77/26

ITEM	RESULT
Prefix / mask	/26 | 255.255.255.192
Addresses per block	2^6 = 64
Network address	192.168.1.64
Broadcast address	192.168.1.127
Usable host range	192.168.1.65 - 192.168.1.126
Usable host count	2^6 - 2 = 62

Formula strip

QUESTION	RULE
Host bits	h = 32 - prefix
Total addresses	2^h
Conventional usable IPv4 hosts	2^h - 2 (network and broadcast reserved in ordinary subnets)
CIDR	/n means n leading network bits; CIDR supports variable-length prefixes.
VLSM	Allocate the largest host requirement first; choose the smallest suitable subnet, then continue with the remaining address space.

Avoid memorising the old classful masks as the default method. The current topic explicitly includes CIDR; solve modern subnetting from the prefix. /31 and /32 have special uses, so 2^h - 2 is not universal for every prefix.

CHAPTER 2 / DATA LINK + NETWORK

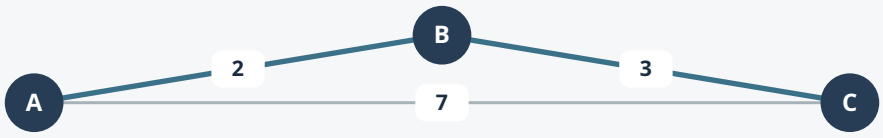
Distance Vector and Link State routing

Both seek routes, but they learn network information differently.

Distance Vector Routing (DVR)

A router maintains a destination cost and next hop, then exchanges distance information with neighbours. It updates routes using the neighbour that gives the lowest total cost.

Update rule: D_x(y) = min over neighbours v of [cost(x,v) + D_v(y)].



Illustrative link costs. Shortest route A to C is A -> B -> C = 5.

ROUTER A DESTINATION	DIRECT COST	BEST COST / NEXT HOP
B	2	2 via B
C	7	5 via B (A -> B -> C = 2 + 3)

Link State Routing: Dijkstra

- 1. Discover neighbours and link costs.
- 2. Flood link-state information so routers build a topology map.
- 3. Run Dijkstra from the source: repeatedly add the nearest unvisited node and relax its outgoing links.
- 4. Build the routing table from each destination's shortest-path cost and next hop.

Same example from A: initialise B = 2 and C = 7; select B at cost 2; update C to min(7, 2 + 3) = 5. Final route to C is through B.

PROTOCOL	ROUTING IDEA	TYPICAL SCOPE
RIP	Distance Vector; hop count, 15 hops maximum for a valid route.	Small/internal networks
OSPF	Link State; shortest-path calculation using link cost.	Within one organisation / AS
BGP	Path Vector; policy and path attributes matter.	Between autonomous systems

Count-to-infinity: routers can keep increasing a bad route after a failure. Split Horizon avoids advertising a route back on the interface it was learned from; Poison Reverse advertises it back as unreachable.

CHAPTER 2 / DATA LINK + NETWORK

MAC access, Ethernet and VLANs

Know how stations share a medium, how collisions happen, and why VLANs are used.

MAC protocol quick table

PROTOCOL	WORKING	KEY LIMIT / POINT
Pure ALOHA	Transmit whenever a frame is ready.	Frames can collide at any time.
Slotted ALOHA	Transmit only at slot boundaries.	Cuts the vulnerable period compared with Pure ALOHA.
1-persistent CSMA	Listen; transmit immediately when idle; keep sensing when busy.	Can cause repeated collisions when channel becomes idle.
Non-persistent CSMA	If busy, wait a random time before sensing again.	Less collision pressure, potentially more delay.
p-persistent CSMA	On an idle slotted channel, transmit with probability p.	Balances waiting and collision risk.
CSMA/CD	Sense, transmit if idle, detect collision, stop and back off.	Classic shared wired Ethernet approach; switched full-duplex Ethernet normally avoids collisions.

Ethernet switching and VLANs

TERM	VIVA-SIZED EXPLANATION
Ethernet frame	A data-link unit containing addressing/control fields and payload; MAC addresses identify local interface endpoints.
Switching	A LAN switch learns source MAC addresses and forwards frames toward the relevant destination port.
VLAN	A logical LAN partition on switching infrastructure. It separates broadcast domains and helps organise access/security. Inter-VLAN communication needs routing.

> Quick distinction: CSMA decides when a device may transmit on a shared medium. VLANs divide a switched LAN into logical network segments.

CHAPTER 3 / TRANSPORT + APPLICATION

TCP, UDP and transport control

Be ready to compare service guarantees and explain who each control mechanism protects.

TCP vs UDP

FEATURE	TCP	UDP
Connection	Connection-oriented; setup before data.	Connectionless; no transport setup.
Reliability / order	Reliable, ordered byte stream using acknowledgements and retransmission.	No built-in guarantee of delivery, order or duplicate suppression.
Control	Flow and congestion control.	Minimal transport mechanism; application may add its own.
Overhead	More protocol state and control.	Smaller, simpler header and handling.
Common fit	Web sessions, file transfer, many reliable data services.	Real-time media, DNS-style queries, lightweight messages when suitable.

Three controls that are easy to mix up

CONTROL	PROTECTS AGAINST	IDEA
Flow control	Receiver being overwhelmed.	Sender adapts to receiver capacity/window.
Congestion control	Network being overloaded.	Sender adapts to congestion signals and network conditions.
Error control	Loss/corruption or missing acknowledgements.	Detect errors and recover where supported, using checks, ACKs and retransmissions.

Stop-and-Wait, Go-Back-N and Selective Repeat

METHOD	WHEN A FRAME IS LOST
Stop-and-Wait	Send one frame, wait for its acknowledgement, then continue. Simple but may leave the link idle.
Go-Back-N	Retransmit the missing frame and subsequent outstanding frames in the sender window. Receiver typically accepts in-order frames.
Selective Repeat	Retransmit only the missing/errored frame(s); receiver can buffer out-of-order frames. More buffering and bookkeeping.

> TCP congestion phases: Slow Start grows the congestion window quickly; Congestion Avoidance grows it more gradually; loss triggers a reduction, with Fast Retransmit/Fast Recovery behaviour depending on the TCP variant.

CHAPTER 3 / TRANSPORT + APPLICATION

DNS, HTTP and HTTPS

These were among the strongest repeated topics in the supplied PYQ analysis.

M.IMP | HTTP behaviour appeared in 9 of 11 papers; DNS appeared in 8 of 11.

DNS: name to address

- DNS is a distributed, hierarchical naming system. It lets applications use domain names instead of memorising IP addresses.
- Typical resolution: client stub -> recursive resolver -> root referral -> relevant top-level domain server -> authoritative server -> answer returned. Caches can shorten or change this path.
- Recursive query: the server contacted is asked to return the final answer or an error. Iterative query: the server returns the best answer it has, often a referral to another server.

HTTP request and response shape

REQUEST

```
GET /api/items HTTP/1.1
Host: example.test
Accept: application/json

[optional body]
```

RESPONSE

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: ...

{"items": [...]}
```

Illustrative format; headers and body depend on the request.

CONCEPT	REMEMBER THIS
Request	Method + path + HTTP version, then headers, blank line, and an optional body.
Response	HTTP version + status code/reason, then headers, blank line, and optional body.
Persistent connection	Reuses a TCP connection for multiple exchanges.
Non-persistent connection	Classic model: a separate TCP connection for each requested object.
Statelessness	HTTP does not inherently remember prior requests. Applications can maintain state using cookies, tokens or server-side sessions.
HTTPS	HTTP protected by TLS, which provides confidentiality and integrity and helps authenticate the server using certificates.

CHAPTER 3 / TRANSPORT + APPLICATION

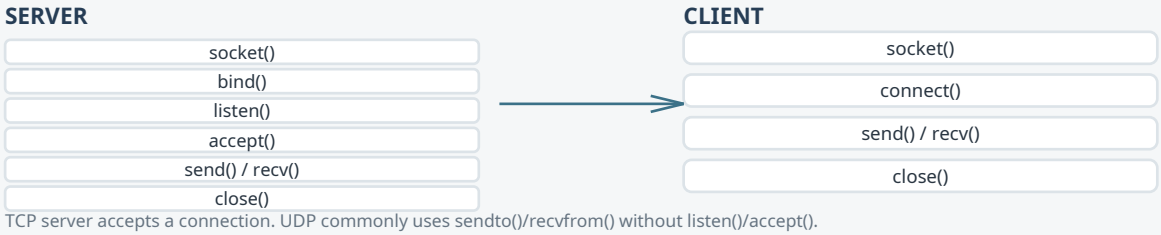
Email, FTP, sockets and REST APIs

Memorise each protocol's job and the order of the client/server calls.

Application protocols

PROTOCOL	ROLE
FTP	Transfers files between a client and server.
SMTP	Sends/submits email between clients/servers and mail servers.
POP3	Retrieves mail, commonly by downloading it to a client.
IMAP	Accesses and synchronises a mailbox while messages generally remain managed on the server.
MIME	Extends email message formats to describe content types and support attachments/non-plain-text content.

Socket programming call flow



TCP server accepts a connection. UDP commonly uses sendto()/recvfrom() without listen()/accept().

socket() creates an endpoint. bind() assigns a local address/port. listen() marks a TCP socket ready for incoming connections. accept() returns a connected socket. connect() initiates a client connection. send()/recv() exchange data; close() releases the socket.

REST APIs and web communication

TERM	SHORT ANSWER
REST API	A resource-oriented web interface commonly using HTTP requests and responses.
HTTP methods	GET retrieves; POST submits/creates; PUT replaces; PATCH partially updates; DELETE removes (actual server behaviour depends on API design).
JSON	A common text representation for structured request/response data. REST is an architectural style, not a transport protocol.

TCP client: socket -> connect -> exchange -> close. TCP server: socket -> bind -> listen -> accept -> exchange -> close. UDP commonly uses sendto/recvfrom and does not require listen/accept.

CHAPTER 4 / WIRELESS, IOT + CLOUD

Wi-Fi and Bluetooth

New-syllabus topics with no historical PYQ coverage in the supplied analysis still belong in viva preparation.

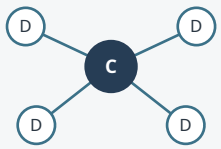
Wi-Fi standards: fast comparison

STANDARD	COMMON BAND	KEY RECALL
802.11b	2.4 GHz	Early Wi-Fi; lower data rate than later generations.
802.11a	5 GHz	Uses 5 GHz; less overlap with crowded 2.4 GHz devices.
802.11g	2.4 GHz	Improved data rate over 802.11b in the same band.
802.11n (Wi-Fi 4)	2.4 and 5 GHz	MIMO support; broader deployment across both bands.
802.11ac (Wi-Fi 5)	5 GHz	Wider channels and multi-user/MIMO enhancements.
802.11ax (Wi-Fi 6)	2.4 and 5 GHz	Designed for efficiency and dense environments; Wi-Fi 6E extends into 6 GHz where supported.

Listed data rates are not shown as guaranteed real-world speed: performance depends on channel width, interference, signal strength, client support and network configuration.

Bluetooth architecture

Piconet



One coordinating device with connected peers

Scatternet



Interconnected piconets; a device may participate in both

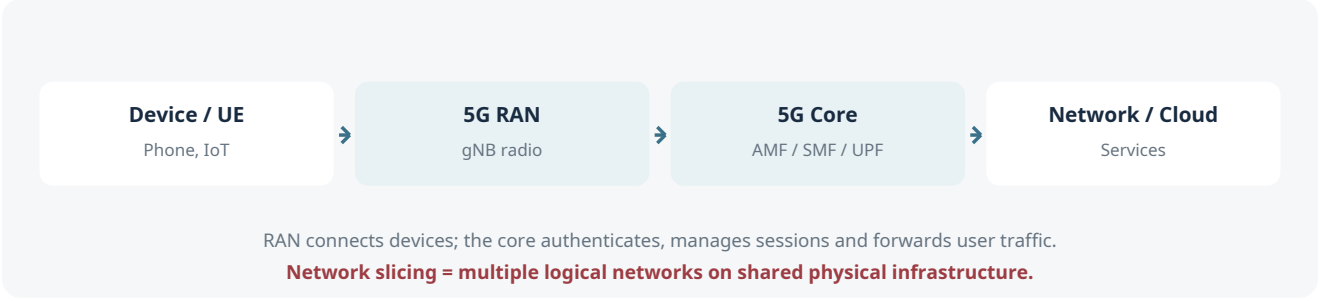
TERM	REMEMBER THIS
Piconet	A Bluetooth group organised around a coordinating device with connected peers.
Scatternet	Two or more piconets connected through a device participating in multiple groups.
Frequency hopping	Bluetooth changes radio frequencies within the 2.4 GHz ISM band to help manage interference and share spectrum.

CHAPTER 4 / WIRELESS, IOT + CLOUD

5G, MQTT and CoAP

Know the main architectural blocks and why IoT protocols keep messages lightweight.

5G architecture



RAN connects devices; the core authenticates, manages sessions and forwards user traffic.

Network slicing = multiple logical networks on shared physical infrastructure.

BLOCK	BASIC ROLE
UE	User equipment, such as a phone or IoT device.
5G RAN / gNB	Radio access network; the gNB provides the radio connection to the device.
5G Core	Core network functions handle access/mobility, sessions and user-plane forwarding. AMF, SMF and UPF are examples of these functions.
Network slicing	Creates separate logical network instances for different service needs over shared infrastructure.

MQTT vs CoAP

FEATURE	MQTT	CoAP
Model	Publish/subscribe through a broker.	REST-like request/response between endpoints.
Transport	Typically TCP; TLS can secure the connection.	Typically UDP; supports compact constrained-device exchanges.
Strength	Many devices can publish to topics consumed by subscribers.	Low-overhead interactions with resource-oriented semantics.
Good fit	Telemetry distribution, sensor updates and event messaging.	Constrained devices and simple machine-to-machine resource access.

MQTT is broker-based publish/subscribe. CoAP is a compact web-style request/response protocol for constrained environments.

CHAPTER 4 / WIRELESS, IOT + CLOUD

Cloud networking and container networking

Keep the service models distinct and explain what the virtual network connects.

Cloud service models

MODEL	WHAT THE USER GETS	NETWORKING ANGLE
IaaS	Virtual machines, storage and networking building blocks.	Virtual networks, routing, security rules and connectivity.
PaaS	A managed platform for building/deploying applications.	Application connectivity and managed service endpoints.
SaaS	A complete application delivered as a service.	Client access, identity and secure data exchange.

Virtual machines vs containers

VIRTUAL MACHINES	CONTAINERS
Each VM runs a guest operating system above a hypervisor.	Containers isolate application processes while sharing the host OS kernel.
Typically heavier; strong separation of guest OS environments.	Typically lighter and faster to start; share the kernel.
Virtual NICs connect VMs to virtual switches/networks.	Virtual bridges/overlays and ports connect containers and services.

Container networking: mental model



Addresses, routing, DNS/service discovery and policies let workloads communicate.

> Cloud networking is about connecting workloads and services while controlling reachability. A VM is not a container, and a container does not normally carry its own separate kernel.

CHAPTER 5 / SECURITY + EMERGING

Firewalls, VPNs, IDS and IPS

Explain what each control observes, protects and does when it sees a problem.

Firewall types

TYPE	HOW IT WORKS	KEY POINT
Packet filtering	Matches packet fields such as IP addresses, ports and protocol against rules.	Simple and efficient; limited application context.
Stateful inspection	Tracks connection state and evaluates packets in that context.	Understands whether traffic belongs to an expected flow.
Application-level gateway	Proxies or inspects application-layer conversations.	More application awareness, usually more processing/configuration.

VPN: private communication over an untrusted network

TYPE	WHAT IT CONNECTS
Remote-access VPN	An individual user/device to an organisation network.
Site-to-site VPN	One network/site to another network/site through a tunnel.

Tunnelling encapsulates traffic across another network; encryption protects confidentiality. IPsec is commonly used at the IP layer. TLS-based VPNs use TLS to protect the tunnel. A VPN does not automatically make every endpoint trustworthy.

IDS vs IPS

IDS	IPS
Detects suspicious activity and raises alerts; often monitors a copy of traffic.	Can be placed inline to block or drop traffic as well as detect it.
May use signatures or anomaly-based detection.	May use signatures or anomaly-based detection; false positives can disrupt legitimate traffic.

> Viva distinction: A firewall enforces traffic policy. An IDS identifies suspicious activity. An IPS can actively prevent detected traffic from proceeding.

CHAPTER 5 / SECURITY + EMERGING

Cryptography and secure communication

Do not mix up encryption, hashing, authentication and digital signatures.

Cryptography essentials

CONCEPT	PURPOSE	EXAMPLE / NOTE
Symmetric encryption	Same secret key encrypts and decrypts.	AES; fast for bulk data, but the key must be shared securely.
Asymmetric cryptography	Uses a public/private key pair.	RSA is a common example; public-key operations can support key establishment or signatures.
Hashing	Maps input to a fixed-length digest; not designed to be decrypted.	Used for integrity checks; a plain hash alone does not prove who sent the data.
Digital signature	Private-key operation verified with the signer's public key.	Supports authenticity and integrity, not confidentiality by itself.

How HTTPS uses TLS

- 1. Client connects to the server and negotiates TLS parameters.
- 2. The server presents a certificate; the client checks that it chains to a trusted issuer and matches the intended server name, within the applicable validity and policy checks.
- 3. The peers establish session keys through the negotiated TLS handshake (exact details vary by TLS version and cipher suite).
- 4. Application HTTP data is protected for confidentiality and integrity by the established TLS session.

> Remember: HTTPS = HTTP protected by TLS. Certificates help authenticate the server; TLS session keys protect the exchange. The handshake is not identical across every TLS version.

Secure communication checklist

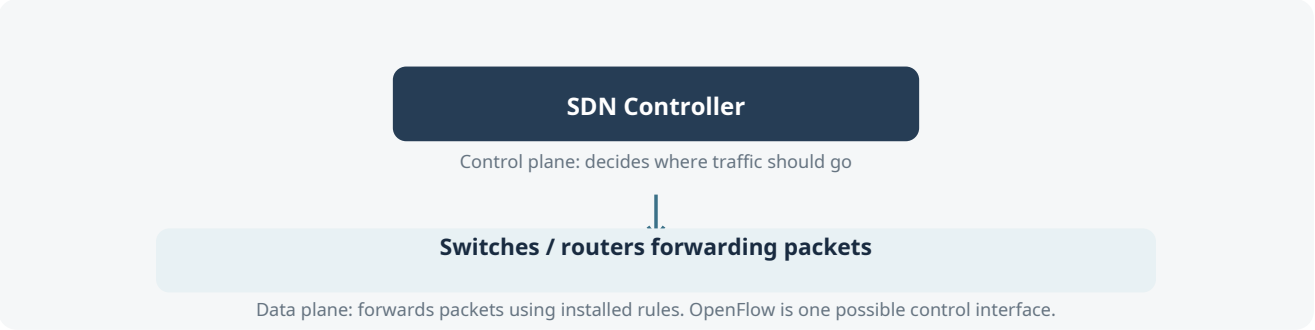
CONTROL	WHY IT MATTERS
Authentication	Establishes which identity or server you are communicating with.
Confidentiality	Prevents unauthorised reading of protected data.
Integrity	Helps detect unauthorised changes to data.
Good practice	Keep systems updated, use least privilege, segment networks, protect credentials and monitor for unusual activity.

CHAPTER 5 / SECURITY + EMERGING

SDN, zero trust and AI-assisted monitoring

These current-syllabus concepts are newer, so keep their purpose and architecture clear.

Software Defined Networking (SDN)



SDN separates the control plane (decides paths and rules) from the data plane (forwards packets). A controller manages forwarding behaviour through a suitable interface; OpenFlow is one example of such an interface.

Zero-trust networking

PRINCIPLE	WHAT IT MEANS
Verify explicitly	Authenticate and authorise based on identity, device/context and policy.
Least privilege	Grant only the access needed, for only as long as needed.
Assume breach	Use segmentation and monitoring so one compromise does not imply unrestricted access.

AI-driven network monitoring

Network telemetry	Feature analysis	Anomaly detection	Alert / response	Human review and feedback

Use cases include spotting unusual traffic patterns, detecting possible failures and helping analysts prioritise alerts. Models can produce false positives and miss unfamiliar behaviour, so human validation, good telemetry and ongoing tuning remain important.

> Viva summary: SDN changes how forwarding is controlled; zero trust changes how access is granted; AI monitoring helps identify unusual network behaviour but does not remove the need for verification.

FINAL PASS / RAPID-FIRE VIVA

One-minute answers

Answer in one or two sentences first. Add the diagram or example only when asked.

What is the difference between OSI and TCP/IP?
OSI is a seven-layer reference model; TCP/IP is a practical protocol architecture commonly presented with four layers.

What is CRC?
A data-link error-detection method using modulo-2 division; the remainder is appended to the data as the codeword.

Distance Vector vs Link State?
Distance Vector exchanges route distances with neighbours. Link State distributes topology information and computes shortest paths.

What does CIDR /26 mean?
26 bits identify the network prefix and 6 bits remain for host addressing.

Why can DVR count to infinity?
Routers can keep trusting stale neighbour routes after a failure. Split Horizon and Poison Reverse help prevent loops.

TCP vs UDP?
TCP provides a reliable ordered byte stream; UDP sends independent datagrams without built-in delivery or ordering guarantees.

Flow vs congestion control?
Flow control protects the receiver. Congestion control protects the network from overload.

Why is HTTP called stateless?
Each request is independent at the protocol level; applications can preserve session state using cookies, tokens or server-side storage.

Recursive vs iterative DNS?
Recursive asks a resolver to return the final answer; iterative accepts referrals and continues querying the next server.

MQTT vs CoAP?
MQTT uses broker-based publish/subscribe. CoAP uses compact REST-like request/response for constrained devices.

VM vs container?
A VM runs its own guest OS above a hypervisor. Containers isolate processes while sharing the host kernel.

IDS vs IPS?
An IDS detects and alerts. An IPS can sit inline and block suspicious traffic.

Encryption vs hashing?
Encryption is reversible with the appropriate key. Hashing produces a digest and is not designed to be decrypted.

What is network slicing?
It creates multiple logical 5G networks on shared physical infrastructure for different service needs.

What is SDN?
SDN separates control decisions from packet forwarding and lets a controller manage forwarding rules.

> Final rule for the viva: define it -> state the problem it solves -> explain the working -> give one practical use or limitation. Keep the first answer short, then let the examiner choose the depth.